Movie Recommendation System

Personalized Viewing for a Next-Gen Streaming Platform

MovieLens Dataset • SVD • TF-IDF • Hybrid Routing • Streamlit

Business Problem

Why does personalization matter?

The Challenge

A new streaming platform needs to compete with Netflix & Disney+.

Without personalization, users churn within the first few sessions.

The Cost of Churn

Acquiring a new subscriber costs 5–7× more than retaining one.

Poor recommendations = lost revenue and brand damage.

Our Solution

A multi-stage recommendation engine that adapts to every user — from first login to long-term resident.

Goal: Increase user retention by delivering the right movie to the right user at the right time.

Dataset: MovieLens — 100,000+ ratings across 9,000+ movies

Methodology: CRISP-DM (Business Understanding → Deployment)

Deployment: Interactive Streamlit dashboard

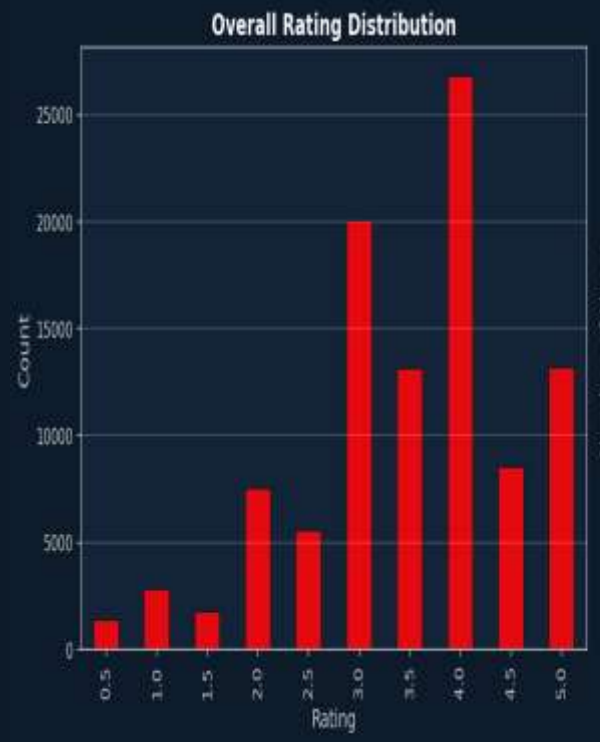
Dataset & Exploratory Data Analysis

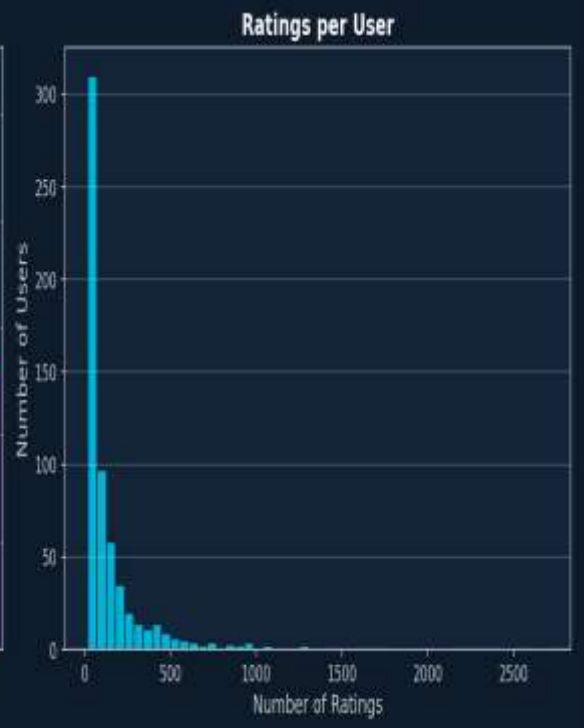
MovieLens 100K

100,000+	9,000+	600+	18
User Ratings	Unique Movies	Active Users	Genre Categories

Key EDA Findings

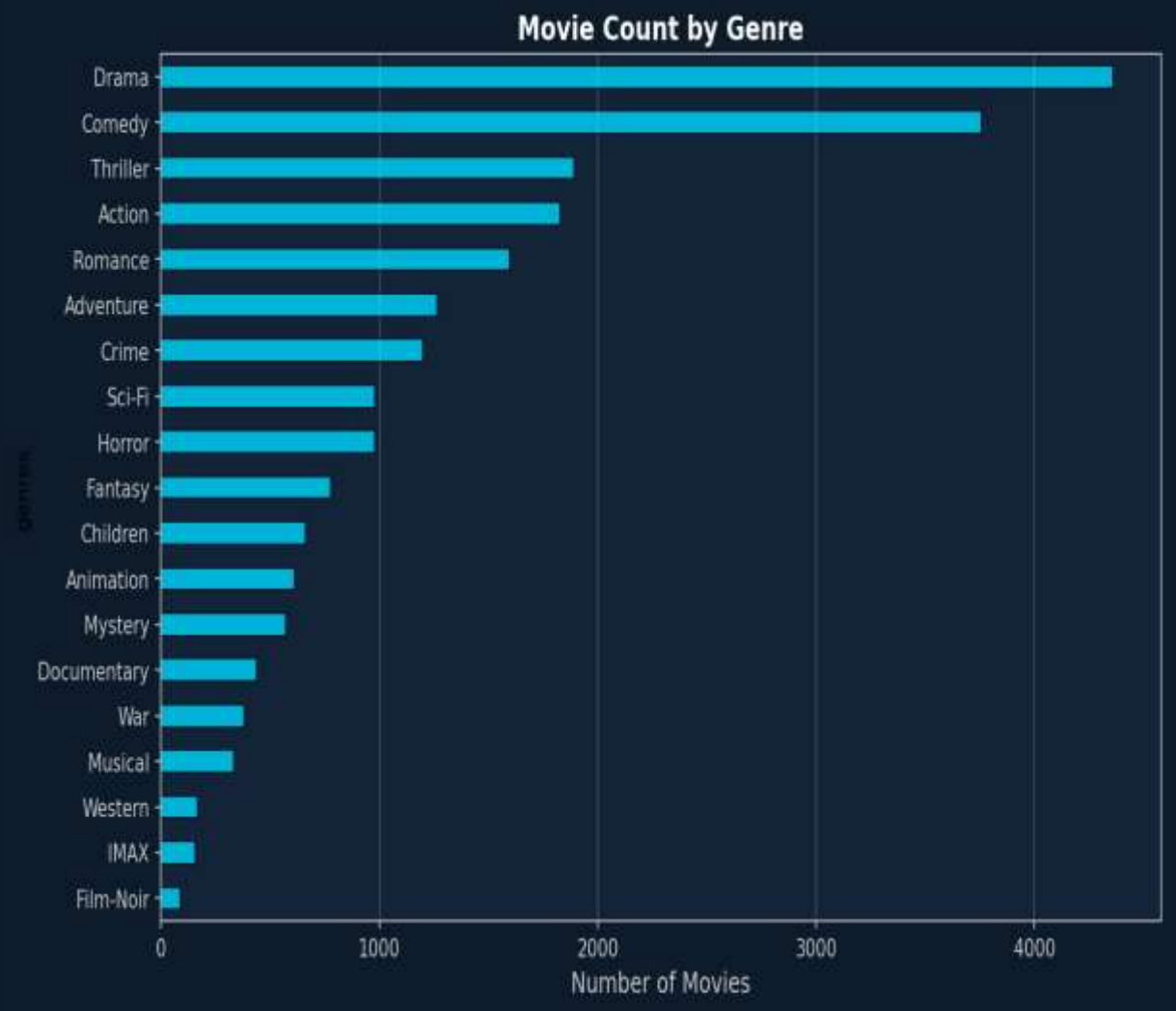
Rating distribution is left-skewed — users tend to rate movies they enjoyed (avg ≈ 3.5)
Drama, Comedy, and Thriller dominate the catalog; niche genres are underrepresented
Power users (top 5%) account for ~30% of all ratings — classic long-tail distribution
~40% of users have fewer than 10 ratings, confirming a significant cold-start challenge
User tags (e.g. 'atmospheric', 'twist ending') provide rich semantic signals for TF-IDF

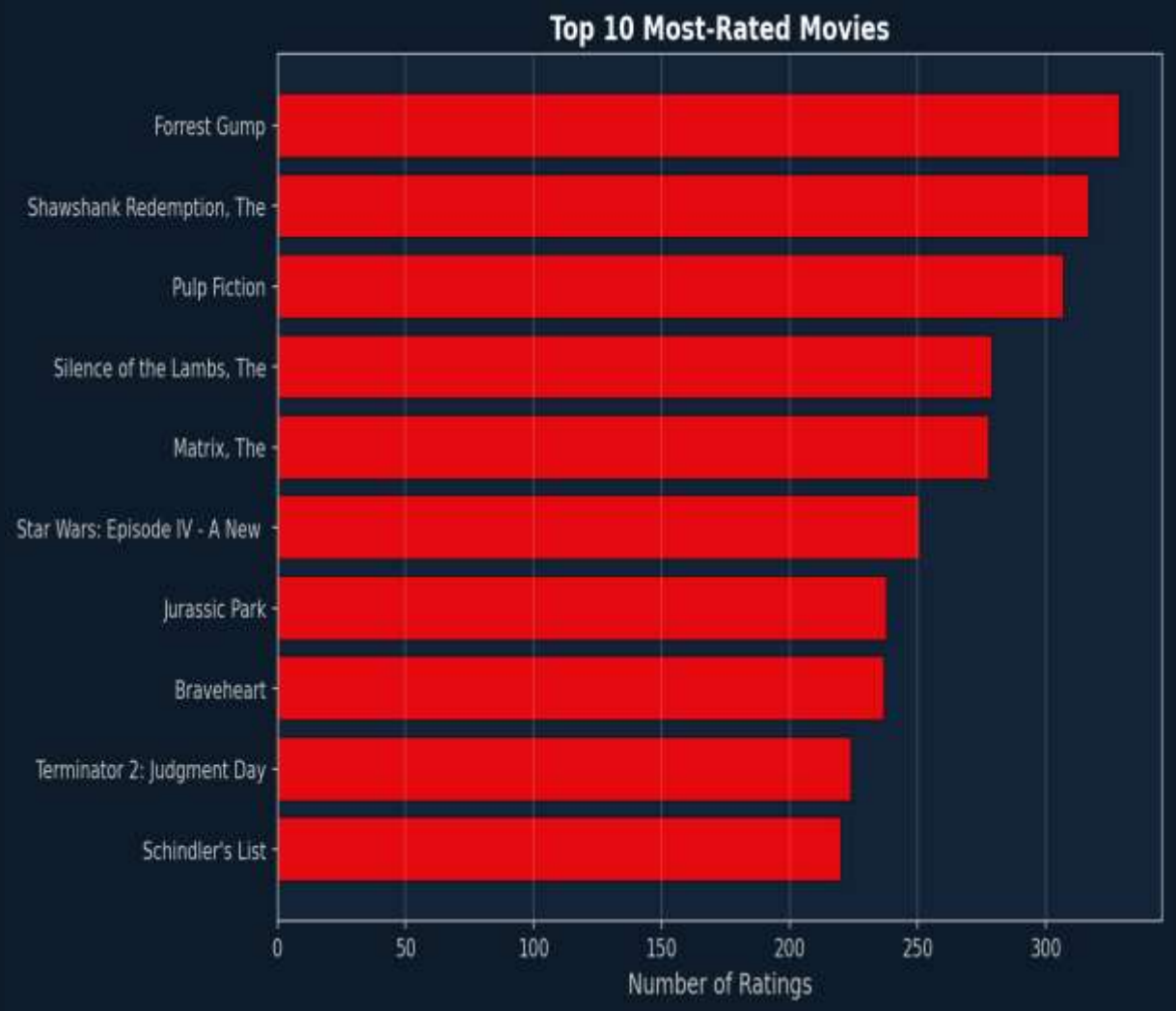




EDA — Visual Insights

Genre distribution & most-rated movies





System Architecture — The Router

One engine, three strategies

User Login → Count Ratings → Router → Select Strategy → Return Recommendations

New / Near-Cold
(0–10 ratings)

Popularity-Based
+
Content-Based (TF-IDF)

Ensures new users see relevant content immediately

Early User
(10–50 ratings)

Hybrid Model
(SVD + TF-IDF, α-weighted)

Balances exploration with personalization

Resident User
(50+ ratings)

Collaborative Filtering
(Optimized SVD)

Full latent-factor personalization

α = 0.7 (SVD weight) in Hybrid mode — tuned to favour collaborative signal while preserving content diversity

Algorithms Deep-Dive

How each model works

SVD
(Collaborative Filtering)

- Decomposes the user-item matrix into latent factors
- Captures hidden taste patterns (e.g. 'likes slow-burn dramas')
- Trained with SGD; regularised to prevent overfitting
- Best RMSE: 0.861

TF-IDF
(Content-Based)

- Builds a feature vector from genres + user tags per movie
- Cosine similarity between user profile and all movies
- User profile = weighted average of rated-movie vectors
- Ideal for cold-start & niche discovery

Hybrid
(α-Weighted Blend)

- Final score = $\alpha \times SVD_norm + (1-\alpha) \times Content_norm$
- $\alpha = 0.7$ (SVD dominates, content adds diversity)
- Explanation engine: labels each rec with a human reason
- Smooth transition from cold-start to full CF

KNN
(User-Based CF)

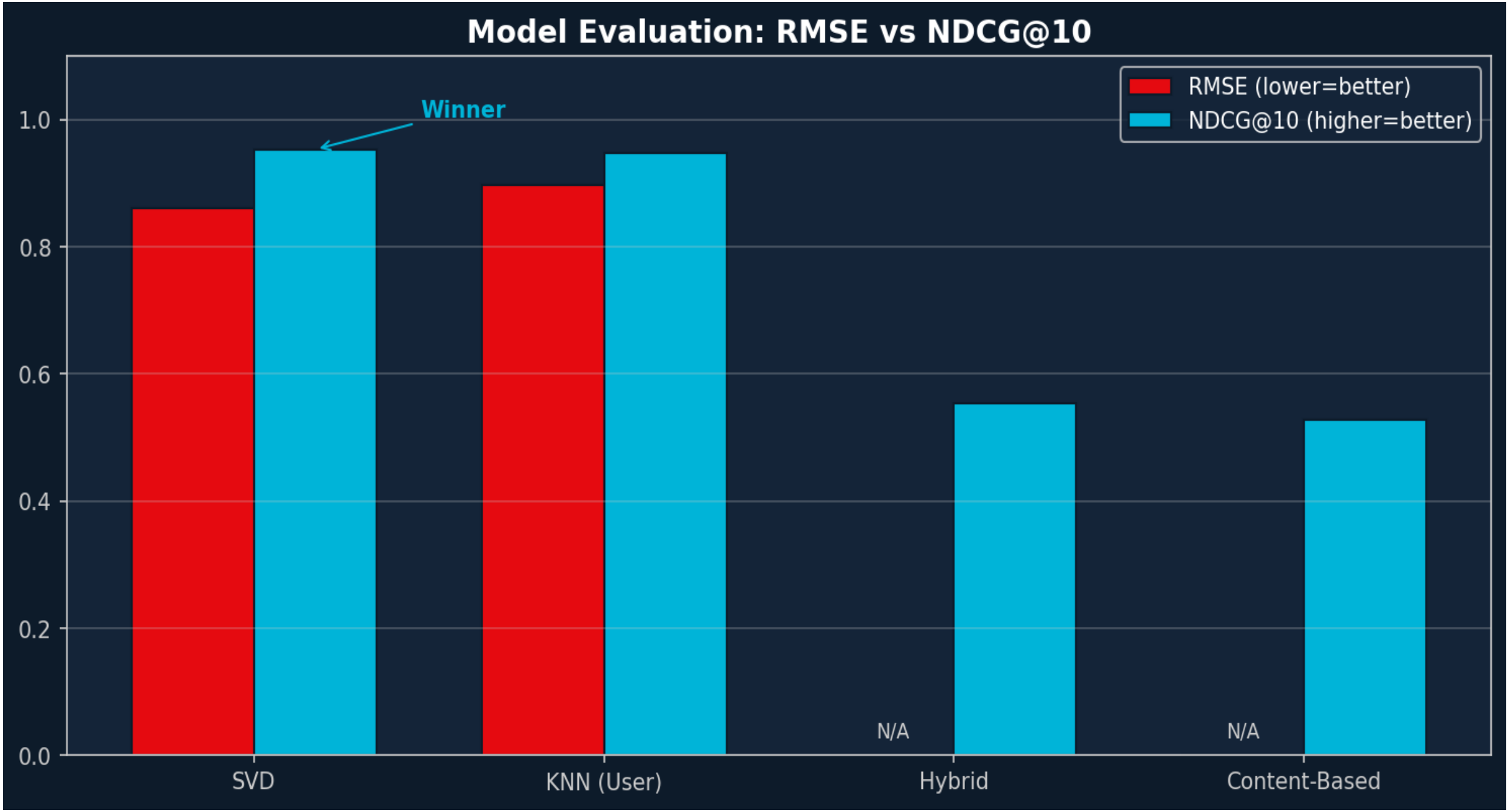
- Finds k nearest neighbours by rating similarity
- Aggregates neighbour ratings for unseen movies
- RMSE: 0.898 — solid but slower than SVD
- Useful for community-based recommendations

Evaluation Results

RMSE (lower = better) • NDCG@10 (higher = better)

Model	RMSE	NDCG@10	Best For
SVD (Collaborative)	**0.861**	**0.953**	**Active / Resident users**
KNN (User-Based CF)	0.898	0.948	Community-driven recs
Hybrid (SVD + TF-IDF)	—	0.554	Early users (10–50 ratings)
Content-Based (TF-IDF)	—	0.527	New users / cold-start

SVD achieves the best RMSE (0.861) and NDCG@10 (0.953) — selected as the primary engine for resident users.


Model Evaluation: RMSE vs NDCG@10
RMSE (lower=better)
NDCG@10 (higher=better)
Winner
1.0
0.8
0.6
0.4
0.2
0.0
N/A
N/A
SVD
KNN (User)
Hybrid
Content-Based

Sample Recommendations

SVD top-5 predictions for selected users

User 1		User 100		User 300	
The Shawshank Redemption (1994)		Lawrence of Arabia (1962)		There Will Be Blood (2007)	
★ 5.0	Crime	★ 4.66	Adventure	★ 4.61	Drama
In the Name of the Father (1993)		There Will Be Blood (2007)		The Shawshank Redemption (1994)	
★ 5.0	Drama	★ 4.62	Drama	★ 4.55	Crime
Dr. Strangelove (1964)		The Shawshank Redemption (1994)		Three Billboards Outside Ebbing (2017)	
★ 5.0	Comedy	★ 4.61	Crime	★ 4.54	Crime
Rear Window (1954)		The Philadelphia Story (1940)		Citizen Kane (1941)	
★ 5.0	Mystery	★ 4.58	Comedy	★ 4.53	Drama
Casablanca (1942)		Dr. Strangelove (1964)		Ghost in the Shell (1995)	
★ 5.0	Drama	★ 4.57	Comedy	★ 4.52	Animation

Streamlit Dashboard

Interactive deployment

User ID Input

Enter any user ID to instantly retrieve personalised recommendations.

Recommendation Slider

Choose between 5 and 20 recommendations per session.

Onboarding Flow

New users rate 10 popular movies to bootstrap their content profile.

Automatic Routing

The router silently selects the best algorithm based on rating history.

Feedback Buttons

Users mark each recommendation as Interested or Not Interested.

Dual-Column Layout

Cold-start users see Popular Picks alongside Content-Based picks side by side.

Run locally: streamlit run app.py | Dependencies: streamlit, pandas, numpy, scikit-learn, scikit-surprise

Conclusions & Next Steps

What we built and where to go next

Key Achievements

- SVD delivers NDCG@10 = 0.953 — near-perfect ranking quality
- Cold-start fully mitigated via popularity + TF-IDF onboarding
- Hybrid model bridges the gap for early-stage users
- Router logic adapts seamlessly across all user lifecycle stages
- End-to-end deployment via Streamlit with real-time feedback loop
- CRISP-DM methodology ensures reproducible, business-aligned results

Future Improvements

- Deep Learning: Neural Collaborative Filtering (NCF) or two-tower model
- Real-time retraining pipeline with streaming user events
- A/B testing framework to measure retention lift in production
- Incorporate implicit feedback (watch time, re-watches, skips)
- Expand content features: cast, director, synopsis embeddings
- Cloud deployment on AWS (SageMaker + Lambda + API Gateway)

Thank You

Group 1 — Movie Recommendation System — Phase 4

Questions?

0.861	0.953	100K+	4
RMSE	NDCG@10	Ratings	Models